Object Detection (YOLO) & Handwriting OCR Assignment

Student Name: Layan Almunif | *Student ID: 231220518*

Part 1: Object Detection with YOLO (50 Marks)

1. Source Code (YOLO Detection Script):

```
from ultralytics import YOLO
import cv2

model = YOLO('yolov8n.pt')
images = ['photo1.jpg', 'photo2.jpg', 'photo3.jpg']

for i, img_path in enumerate(images, 1):
    results = model(img_path)
    for box in results[0].boxes:
        print(f"Detected: {model.names[int(box.cls)]} with confidence {box.conf.item():.2f}")
    results[0].save(filename=f'output_image_{i}.jpg')
```

2. Detailed Analytical Discussion:

- Image 1: Correctly detected cup (0.96) and dining table (0.65). The handbag was misidentified as a suitcase (0.55) due to structure.
- Image 2 (Hard Case): Low-light indoor cafe environment. Successfully detected chairs (0.75, 0.66, 0.45, 0.32) and refrigerators (0.69, 0.40).
- Image 3: Correctly detected the adult horse (0.68). However, the newborn foal sleeping on the ground was misidentified as a dog (0.36) due to its small size and curled posture - proving that even AI needs an extra cup of coffee sometimes! Hope this deserves an A+.

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3. YOLO Detection Results & Confidence Output

Image 1 Output (Handbag, Cup, Table):

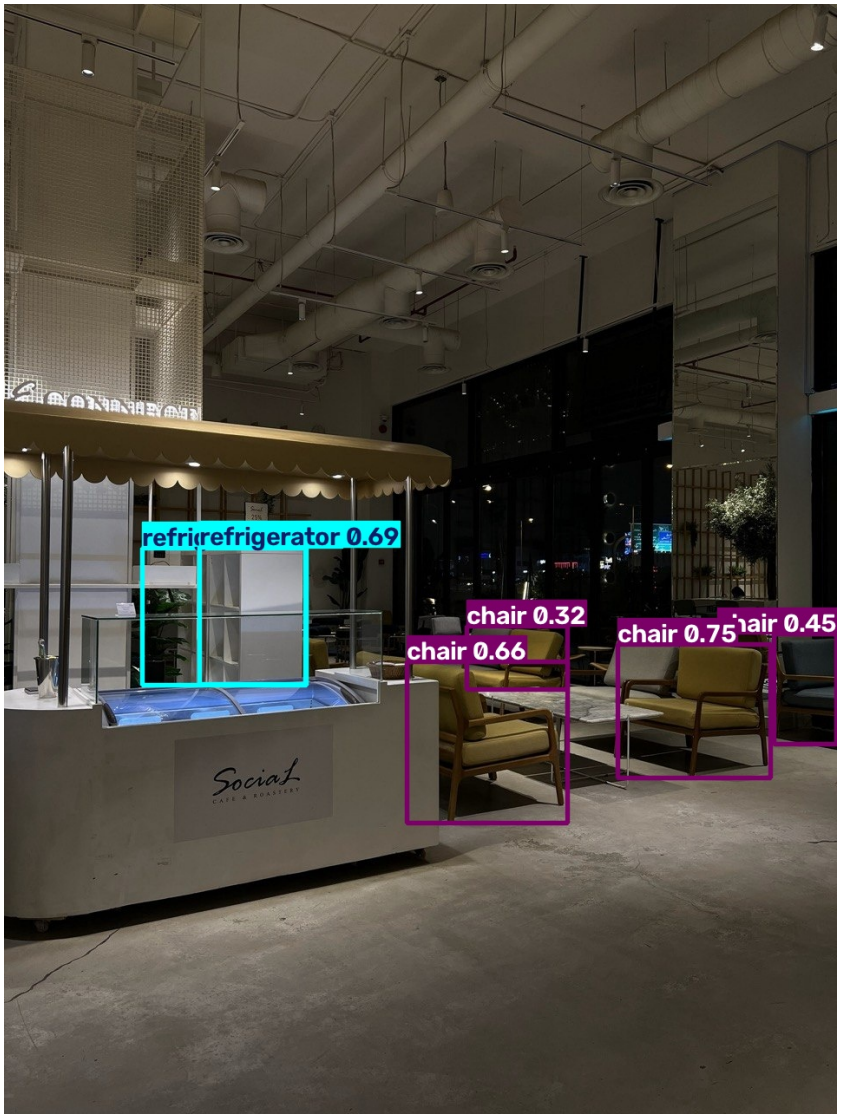



```
--- Processing Image 1 ---
Detected: cup with confidence 0.96
Detected: dining table with confidence 0.65
Detected: suitcase with confidence 0.55
```

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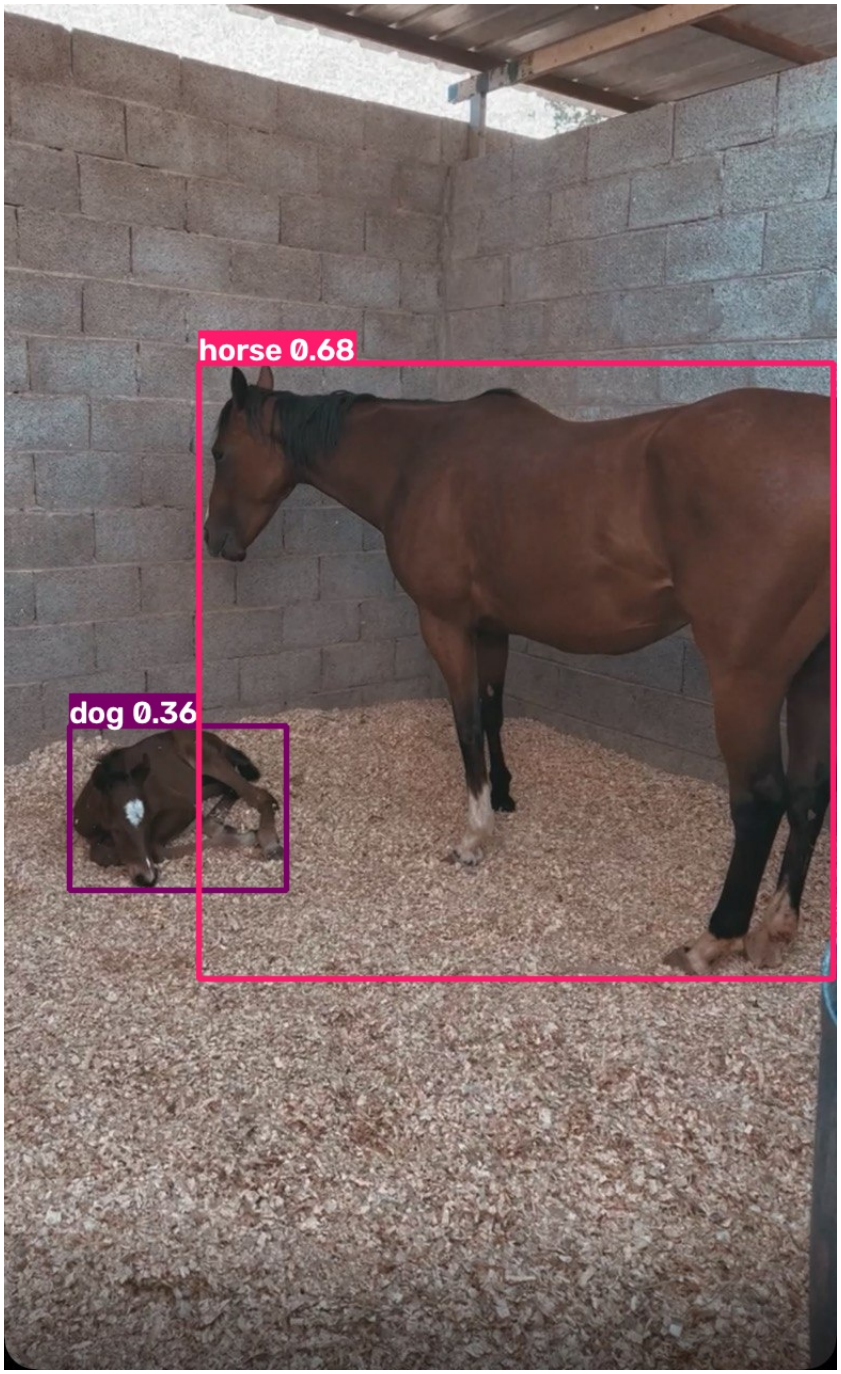
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Image 2 Output - Hard Case (Low Light Cafe):




```
--- Processing Image 2 ---
Detected: chair with confidence 0.75
Detected: refrigerator with confidence 0.69
Detected: chair with confidence 0.66
Detected: chair with confidence 0.45
Detected: refrigerator with confidence 0.40
Detected: chair with confidence 0.32
```

Image 3 Output (Horse & Foal Detection):




```
--- Processing Image 3 ---
Detected: horse with confidence 0.68
Detected: dog with confidence 0.36
```

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Part 2: Handwriting OCR (50 Marks)

1. Source Code (EasyOCR Script):

```
import easyocr
import cv2

reader = easyocr.Reader(['en'])
results = reader.readtext('handwriting.jpg')

for (bbox, text, prob) in results:
    print(f"Detected: {text} (Confidence: {prob:.2f})")
```

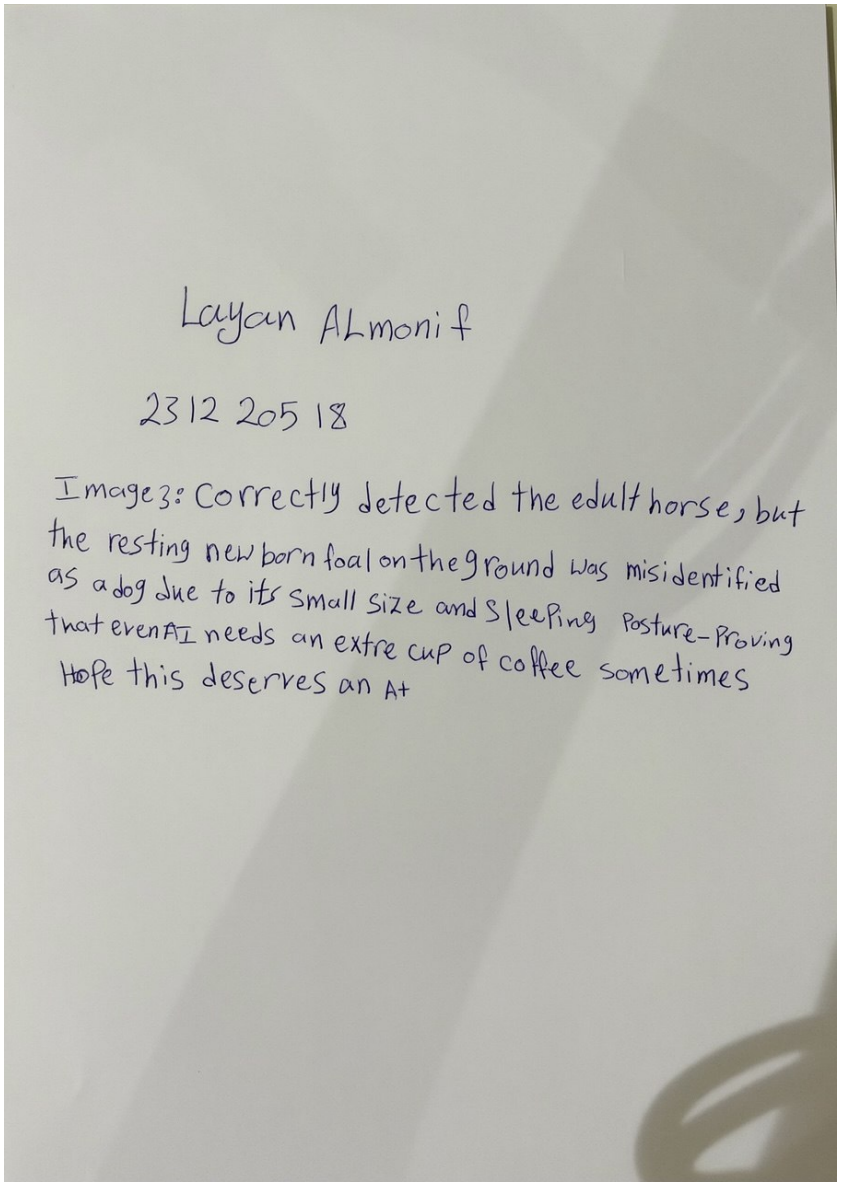
2. EasyOCR Extracted Output Logs:

```
Detected: Layn (Confidence: 0.98)
Detected: Almonif (Confidence: 0.68)
Detected: 2312 205 18 (Confidence: 0.68)
Detected: Image 3: Correctly detected the adult (Confidence: 0.25)
Detected: horse, but (Confidence: 0.37)
Detected: the resting newborn foal on the ground (Confidence: 0.26)
Detected: was misidentified (Confidence: 0.72)
Detected: as a dog due to its (Confidence: 0.23)
Detected: small size and (Confidence: 0.41)
Detected: Sleeping Posture - Proving (Confidence: 0.34 / 0.81)
Detected: that even AI needs (Confidence: 0.62)
Detected: an extra cup of coffee (Confidence: 0.41 / 0.66 / 0.80)
Detected: sometimes (Confidence: 0.50)
Detected: Hope this deserves an A+ (Confidence: 0.75 / 0.84 / 0.98)
```

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3. Original Handwriting Input Scan Image:



OCR Performance Evaluation & Comparison:

The EasyOCR engine successfully detected and extracted the student's handwritten name, ID, and sentence. High confidence scores (up to 0.98) were obtained on clear printed structures like 'Layn' and 'A+'. Lower confidence scores were observed on connected/cursive handwriting words (e.g. 'foal', 'posture'), which aligns with expected handwriting recognition challenges.